MIL-R-46453D (PA)
14 February 1975
SUPERSEDING
MIL-R-46453C (MU)
31 January 1967
MIL-R-46447C
25 January 1967
(see 6.4)

MILITARY SPECIFICATION

RELAY, M7, PARTS FOR LOADING, ASSEMBLING AND PACKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 This specification covers the parts for, loading, assembling and packing for one type of relay designated as Relay, M7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

*MILITARY

MIL-P-116 - Preservation, Packaging, Methods of
MIL-A-2550 - Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

*U.S. ARMY PICATINNY ARSENAL

8840608 - Relay, M7, Loading Assembly
9258182 - Box, Set Up, Packing, Ammunition, Inner for Small Explosive Components
9258183 - Box, Set Up, Packing, Ammunition, Outer for Small Explosive Components
9258184 - Box, Packing, Ammunition for Small Explosive Components

(Copies of specifications, standards and drawings required by supplies in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*For meaning of asterisk (*) see paragraph 6.5.

3. REQUIREMENTS

applicable drawings and specifications.

3.2 Relay. - The relay shall comply with all requirements specified on Drawing (dwg.) 8840608, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide. - The moisture content of the lead azide used in loading the relay shall not exceed 0.50 percent.

3.4 Workmanship. - All parts shall be fabricated and loaded in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease, and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used

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shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. (All packing components shall be dry).

*3.5 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be sions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

*4.2 First Article Inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of 25 complete sets of parts (i.e. 25 each of every component) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection. - If any component fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

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4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 Inspection Level II except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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*4.3.2.1 Cup, relay (see dwg. 8840625 covering a detail of dwg. 8840608).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Cup split or cracked	Visual	01001
Major:	AQL 0.65 percent		
101.	Outside diameter, maximum (max.)	Gage	01002
102.	Inside diameter, minimum (min.)	Gage	01003
103.	Thickness of bottom, max.	Gage	01004
Minor:	AQL 1.5 percent		
201.	Total length	Gage	01005
202.	Radius missing	Visual	01006
203.	Evidence of poor workmanship (see 3.4)	Visual	01007

*4.3.2.2 Washer, relay (see dwg. 8844437 covering a detail of dwg. 8840608).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Thickness	Gage	02001
102.	Inside diameter	Gage	02002
103.	Outside diameter	Gage	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02004

4.3.2.3 Relay, M7, Loading Assembly (see dwg. 8840608).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cracking or flaking of charge ..	Visual	03001
2.	Cup cracked or split	Visual	03002

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Major:	AQL 0.40 percent		
101.	Charge missing	Visual	03003
Minor:	AQL 2.5 percent		
201.	Outside diameter	Gage	03004
202.	Total length	Gage	03005
203.	Underside of crimped metal fails to contact washer a full 360 degrees	Visual	03006
204.	Any portion of washer hole obstructed by crimp	Visual	03007
205.	Component missing	Visual	03008
206.	Evidence of poor workmanship (see 3.4)	Visual	03009

*4.3.2.4 Box, set-up, inner, after loading and prior to closing slide cover (see dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Filler and cushion assembly missing	Visual	04001
2.	Filler and cushion assembly improperly assembled	Visual	04002
3.	Stiffener missing	Visual	04003
4.	Slide cover assembled reversed ...	Visual	04004
Major:	AQL 0.40 percent		
101.	Release paper missing	Visual	04005
Minor:	AQL 1.00 percent		
201.	Marking missing, incomplete or illegible	Visual	04006
202.	Evidence of poor workmanship (see 3.4)	Visual	04007

*4.3.2.5 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top and sealing strip (see dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.

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Critical:
1. Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box ..... Visual 05001
2. Stiffener not facing one direction Visual 05002

Major: None defined

Minor: AQL 1.00 percent
201. Marking missing, incomplete or illegible ..... Visual 05003
202. Evidence of poor workmanship (see 3.4) ..... Visual 05004

4.3.2.6 Sealed packed barrier bag, prior to packing in box (see dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal	Visual	06001
102.	Evidence of crushed box	Visual	06002

NOTE: Package should be opened if there is evidence of crushing.

Minor:	AQL 1.50 percent		
201.	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied	Visual	06003
202.	Marking of bag, misleading or unidentifiable	Visual	06004
203.	Evidence of poor workmanship (see 3.4)	Visual	06005

*4.3.2.7 Box, packing, ammunition prior to assembling cover and strapping (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical: NOne defined			

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Major:	AQL 0.65 percent		
101.	Bags containing set-up boxes damaged	Visual	07001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	07002
103.	Marking missing, misleading or unidentifiable	Visual	07003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	07004

*4.3.2.8 Box, packing, ammunition (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken, loose or improperly fastened	Visual/ Manual	08001
102.	Box damaged	Visual	08002
103.	Marking missing, incomplete or illegible	Visual	08003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	08004

4.3.3 Testing

4.3.3.1 Moisture content of lead azide (see 3.3) Major Defect, Code No. 09001. - The moisture content of each lot shall be determined at the time of loading. A sample in sufficient quantity to perform the test detailed in 4.4.1 shall be selected and tested as specified in 4.4.1. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all realys loaded with the questionable lead azide shall be rejected (Destructive).

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4.3.3.2 Weight of charge (see dwg. 8840608), Major Defect, Code No. 10001. - One hundred and twenty-five (125) loaded relays shall be randomly selected from each lot for this test. If more than one relay fails to meet the requirements, the lot shall be rejected. Testing shall be as specified in 4.3.2. (Non-destructive).

*4.3.3.3 Heat seal test of sealed bag (see dwg. 9258183), Major defect, Code No. 06006. - Sampling acceptance criteria and test procedure for this test shall be as specified in Specification MIL-P-116. (Destructive)

4.3.4 Inspection Equipment.-Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection design and approval.

4.4 Test Methods and Procedures

4.4.1 Moisture content of lead azide. - The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.6, except that a two (2) gram sample shall be used.

4.4.2 Determination of charge weight. - One hundred pairs of cups and washers shall be weighed prior to loading to estimate the mean ($\overline{X}$) and standard deviation (S) of the cup and washer weights. To compute $\overline{X}$ and S use the following:

$$\overline{X} = \sum X/100 \qquad \overline{R} = R/20$$

$$S = 0.430\ \overline{R}$$

where:

X = weight of each pair of cup and washer
R = range of weights for each group of five pairs of cups and washers (twenty groups)

Next, compute the lower limit (L) for the weight of an individual pair of cup and washer:

$$L = \overline{X} - 2.38S$$

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Next, the specified minimum charge weight shall be added to the lower limit of the weight of cup and washer. The weight so determined shall be the minimum weight of the loaded relay. The weight of each loaded relay in the sample shall be compared to the minimum relay weight. Test equipment shall be as specified in 4.3.4.

*5. PREPRATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C. - Minimum Military Package. - The relay shall be packaged and marked in accordance with dwg. 9258182 and 9258183.

5.1.2 Level C Metal parts. - Metal parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing

5.2.1 Level C. - The relays shall be packed as specified in dwg. 9258184.

5.2.2 Level C Metal Parts. - Metal parts shall be packed in accordance with MIL-STD-1169 requirements.

5.3 Marking

5.3.1 Marking. - Packaging and packing containers for the loaded relay shall be marked in accordance with dwgs. 9258182, 9258183 and 9258184 requirements. Metal parts packaging and packing containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article sample

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

*6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form

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(per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Supersession data. - This specification includes the requirements of MIL-R-46447C (MU), dated 25 January 1967.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
Army-PA

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A. GIVE PARAGRAPH NUMBER AND WORDING

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